

INTRODUCTION TO COMPUTATIONAL METHODS

As it is known, the closed form solution of a Quadratic equation is,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To solve $f(x) = ax^2 + bx + c = 0$

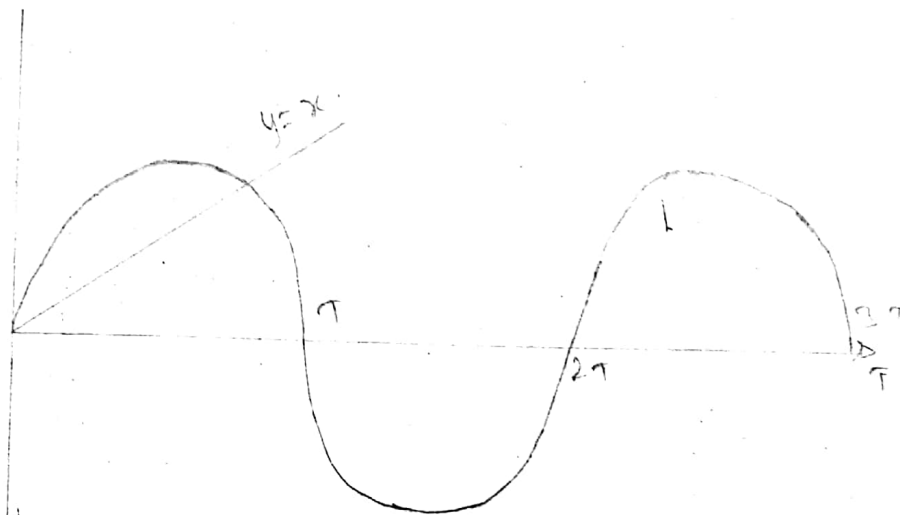
The values obtained are called roots. They represent the values of x that makes equation equal to 0. Thus, we can define the root of an equation as the value of x that makes $f(x) = 0$. Although, this is too easy and there are many other functions for which the root cannot be determined so easily. From these cases the numerical method are used.

Actually there are 3 methods for solving eqⁿ of the form $f(x) = 0$.

- (i) Analytic Method
- (ii) Graphical Method
- (iii) Numerical Method.

(i) Analytic Method: It is not possible to form an analytic expression for the root for every functions. From A polynomial function $f P_n(x)$ of degree n has form $P_n(x) = q_0 + q_1x + q_2x^2 + \dots + q_nx^n$.

Graphical method



To solve,

$$P(x) = ax^2 + bx + c$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Unfortunately such analytic formula do not exist for polynomial of degree 5 or higher. Now there exist two other methods.

(ii) Graphical Method:- A graphical approach is used to find an approximate solution to a problem by viewing and interpreting a graphical image accordingly. In this method, given a transcendental equation in the form $f(x) = 0$. We separate it into two sets of equations in form $f(x) = 0$. We separate it into two sets of equations which can be easily plotted for example. If the equation is $f(x) = x + \sin x = 0$, the two sets are $y = +\sin x$, $y = -x \Rightarrow x = -\sin x$. Now, we plot these two equations on a graph and the root of the equation is the point of intersection of these curves.

(iii) Numerical Method:- A numerical approach is used to find an approximate solution to a problem but may be simpler than an analytical approach. In this method the value of a single root is formed on the basis of first

knowledge of its approximate value. This is based on the following theorem.

If $f(x)$ is continuous in the interval (a, b) & if $f(a)$ and $f(b)$ are of opposite sign, then the equation $f(x) = 0$ will have at least one root between a and b .

Numerical methods are divided into two parts:-

- (a) Bracketing method
- (b) Open method.

(a) Bracketing method:- These methods explain the fact that a function typically changes its sign in the vicinity of a root. They start with two initial guesses that bracket the root and then symmetrically reduce the width of the bracket until the solution to a desired accuracy.

Bracketing methods are:-

- (a) Bisection Method
- (b) False position
- (c) Regula falsi Method
- (d) Improved - Regula falsi Method.

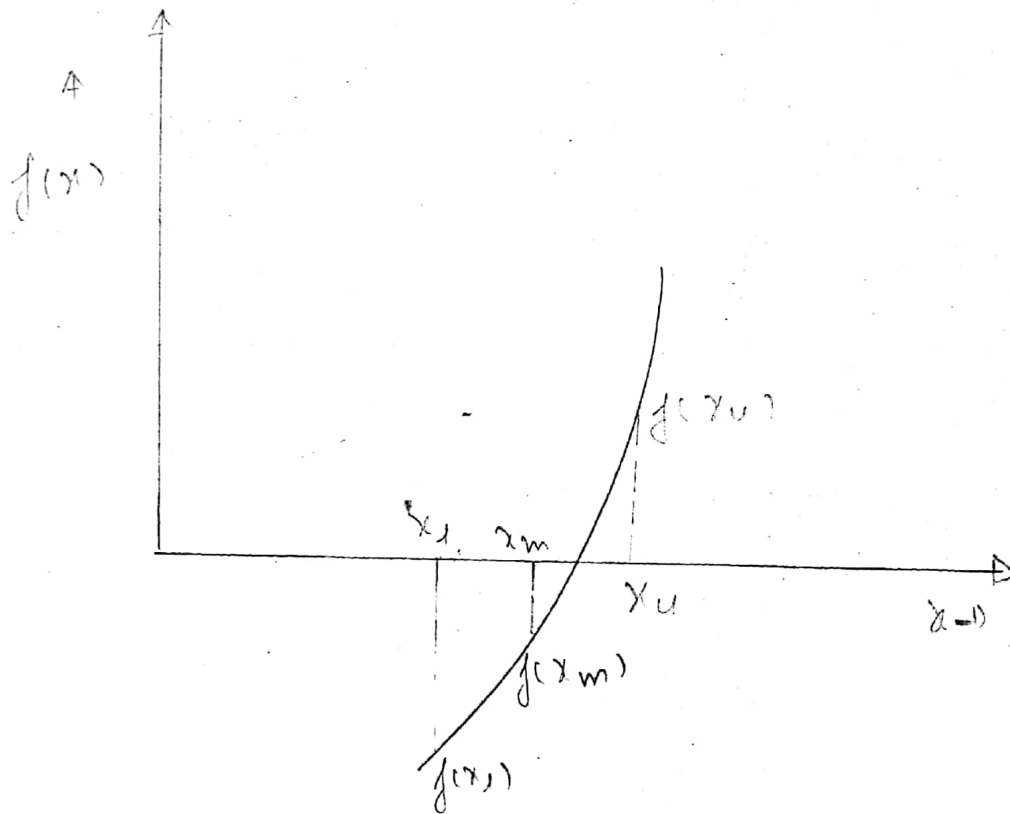
(b) open Methods:- These methods are based on the method that it require only a single starting (or guess) value of solution or two starting values that do not necessarily bracket the root. They may sometime diverge or move away from the root as the computation progresses. However, when the open methods converge, they do ~~so~~ move quickly than the Bracketing method. Some of the popular open methods are:-

- (a) Secant Method
- (b) Newton-Raphson Method
- (c) Bairstow's method
- (d) Muller's method.

Given below are the methods used to find the roots of polynomials.

Bisection Method:- The bisection method in mathematics is a root finding method that repeatedly bisects an interval and then selects a sub-interval in which a root must lie for further processing. It is a simple method, but also slow, because of this it is often used to obtain a rough approximation to a

BISECTION METHOD



Solution which is then used as a starting point for more rapidly converging methods. This method is also called interval halving method, the binary search method; let, bisection method be used to find the root of the polynomial.

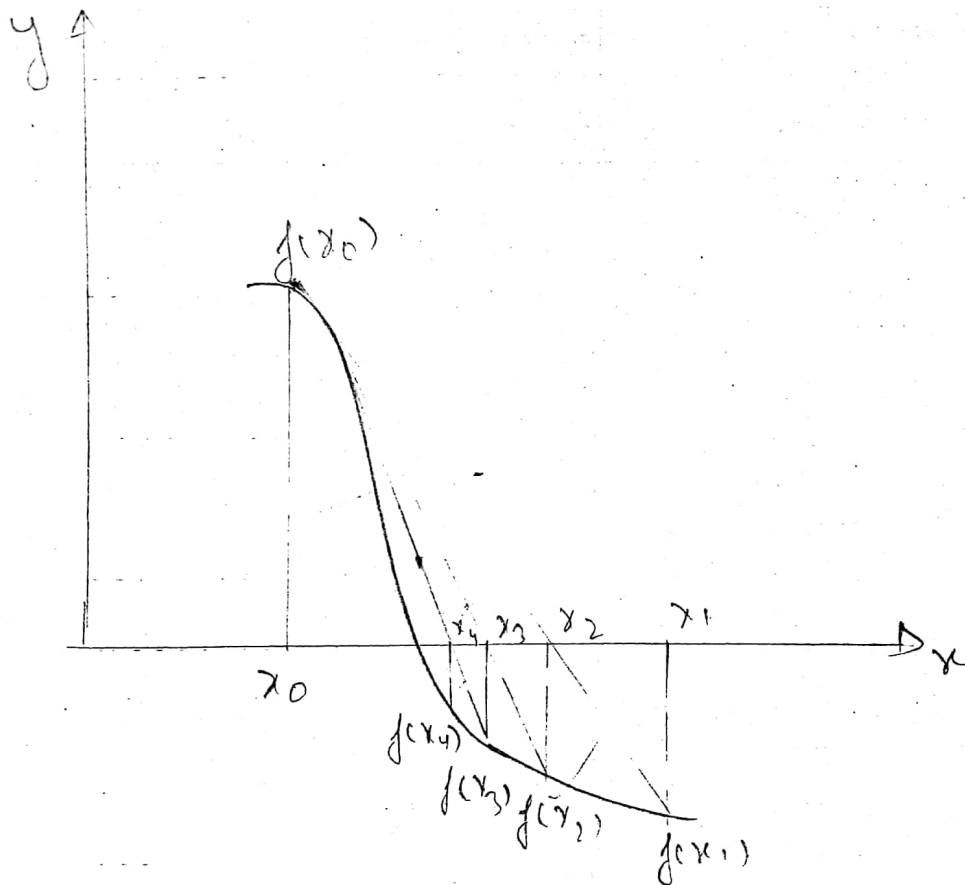
$$f(x) = x^3 - x - x$$

First, two no. a and b have to be found such that $f(a)$ & $f(b)$ have opposite signs. For the above function, $a = 1$ and $b = 2$ satisfy this criterion as

$$f(1) = -2 \quad \text{and} \quad f(2) = 4$$

As, the function is continuous there must be a root within the interval $[1, 2]$.

In the 1st iteration, the end points of the intervals which bracket the root are $a_1 = 1$ and $b_1 = 2$. So, the mid point is 1.5 . The function value at that point is $f(c_1) = f(1.5) = -0.125$. Because $f(c_1)$ is negative, $a = 1$ is replaced with $a = 1.5$ for the next iteration to ensure that $f(a)$ & $f(b)$ have opposite signs. As this continues, the interval between a & b will become increasingly smaller, converging on the root of the function. After a few iterations,



it becomes apparent that there is convergence to about 1.521, a root for the polynomial.

Regula Falsi Method

This is a better method than the bisection method. Here point c is found out by taking the straight line joining the points $(a, f(a))$ and $(b, f(b))$ intersecting the x -axis. To obtain the value of c , we can equate the two expressions of the slope ' m ' of the line ' L '.

$$m = \frac{f(b) - f(a)}{(b-a)} = \frac{0 - f(b)}{(c-b)}$$

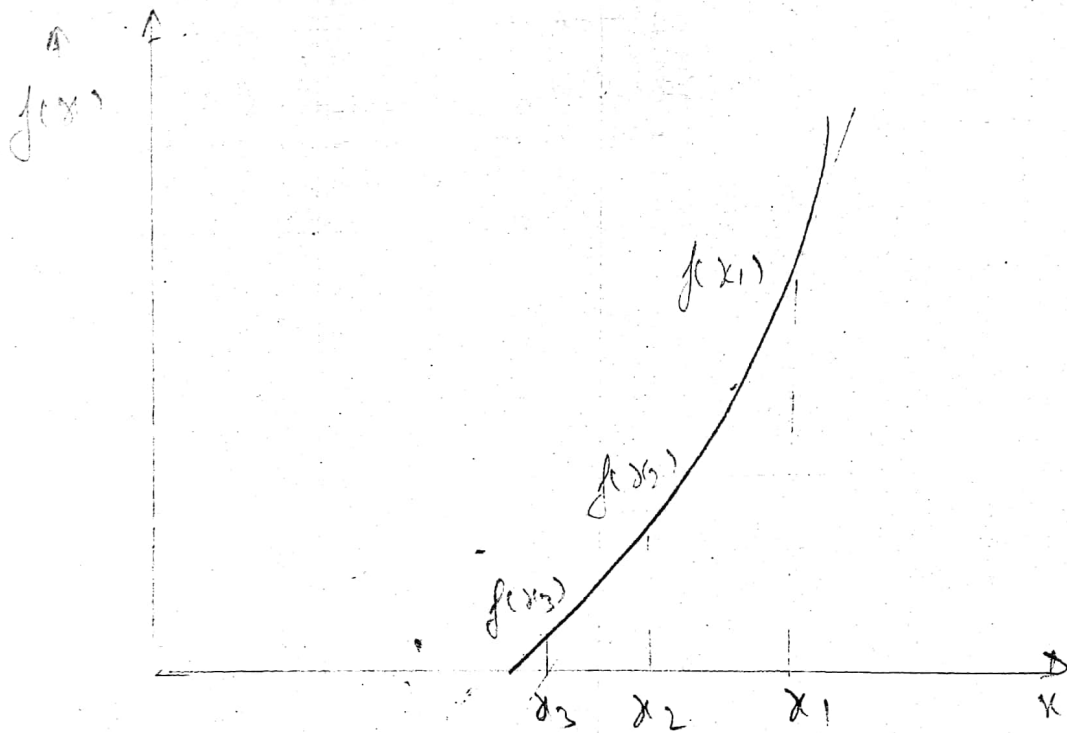
$$\Rightarrow c = b - \frac{f(b) \times (b-a)}{f(b) - f(a)}$$

If $f(b) \times f(c) < 0$, then $b = c$
else $a = c$, or

if b is equal to 0, then c is the root.

Newton - Raphson Method

The underlying idea, used for by Newton and Raphson in this method is the approximation of the graph



of the function $f(x)$ by the longest line.

Let, x be a root of $f(x)$ i.e. $f(x) = 0$. Assume that $f(x) \neq 0$. Let x_1 be a no. close to x . The tangent line to the graph of $f(x)$ at x_1 has x_2 as its intercept.

From the above picture, we see that x_2 is getting close to x . Therefore,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

We continue this

process, getting value of x_n

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

Successive Iteration

In computational mathematics, an iterative method is a mathematical procedure that generates a sequence of improving approximate solutions for a class of problems.

To compute the value of root of $f(x) = 0$, we separate x from the function and express as $x = g(x)$.

For a particular

value of x_1 , we compute $g(x_1)$ and store it as x_2 , which is then used to compute $g(x_2)$ & so on.

Hence,

$$x_{n+1} = g(x_n).$$

As the iteration proceeds, x_n approaches the value of the root of $f(x)$.

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